

Due: April 29th, 2026

Title: Automated HMMER suite search tool for ease of use in Deep Homology Searches-
Ancyromonad 6mA analysis

Background: Hidden Markov Models are useful for finding homologous proteins in species which may be divergent and hard to find in a standard BLAST search. The proteasome being used in this analysis is an *Ancyromonas* proteasome. *Ancyromonas* are deep branching unicellular eukaryotes and their proteins tend to be divergent from model eukaryotes. I will be analyzing whether *Ancyromonas sigmoides* has 6mA methyltransferase-like enzymes similar to other unicellular eukaryotes. HMMER suite tools are used in deep homology searches but take time to install correct packages, switch through tools, and ensure files are all in the correct folder. The output from HMMER can also be hard to read and does not include the sequence, which then has to be extracted from sequence data one by one, for each hit. This pipeline “automates” this process and creates an easy to read table, which pulls and provides sequences.

Methods: A multiple sequence alignment was done to create a “baseline” for the 6mA methyltransferases from ciliate and human 6mA methyltransferases and saved as a .faa file. Ancyromonas proteasome was downloaded as a .faa file. Both .faa files were uploaded to github for use in analysis. The code, which is written in shell, was written as a nano in terminal. The script first prints users working directory to ensure they are in the correct directory, same as their files. Then the script runs a ask_yes_no function to ask if conda and hmmer need to be installed, it skips the installation if the user says “no”. The script continues and gathers the correct files for the HMMbuild function. It asks the user if they have a MSA file, if yes then asks the user to input a name for the output file ending in .hmm. The 6mA multiple sequence alignment (MSA) is input in this step. Then script prompts the user to input the MSA.faa file and outputs the .hmm file. If the correct input is not put in, then echo read “please run a multiple sequence alignment). The script continues to HMMsearch, which compares the 6mA MSA to the *Ancyromonas* proteasome. The script then asks the user to input a proteasome to search, the ancyromonas.faa file is input at this step. The .hmm created in the last step is input again. The user is asked to enter a name for the output file ending in .txt. Script takes inputs, runs search from 6mA MSA.faa input against ancyromonas.faa input and outputs HMMsearch results as .txt file. The user is asked if they want the matching sequences found to be exported to a .xlsx file. 6mA .txt file sequence hits are used to pull E-value, score, bias, FASTA header, and sequence data and organize it into a .xlsx, which can be easily read.

Results:

Sequence ID	E-value	Score	Bias	Description	FASTA Header	Sequence	Sequence Length
Asigmoides_jg3005.t1	1.1e-05	23.5	0.0	DNA binding protein, used in 6mA methylation, RMSD score of 1.746	Asigmoides_jg3005.t1	MVPLPMAQAVTGDGSSAPAKAAPSPADSVRSOPANVSTEDARSAGSSSE ETLGGGAGGPMAMVDVAGAGPSAGGAGSPSPAGSSSACGEGSSSSAA SPSSAASPAGAGSPSPAGSGGAGSGSGSGSGVQAGPSGSGSAEG AAGPSSGAGAGSKRGRAPVPAPTRVQSKRSKTFGMDAIQHALFDGSVT SAGALSGKKKPKAKKKKSAVKKPSAAHYALYATYDEPIEFVMAFQMVVE MLAESREAAAAASTSPAGAAADGNGDGGNGDGSAPAAAAAEAVV DRKGKRAVDPPDDEAAAAACDVPVTLTEKLEILFKTTFKPLSIIKEGEV KAGDGKDVYIENDDAMVSSSDGWSDDSSGERERRAAAKSKRRRREP PREPRRRPRKERAPKRERAAPVIERMVTSQLRDESGRIVLKKSVKFRDPTLP SLTRAPEGTGAELRKSARLIEFEFERDLPQLHADVRVTDLLTYDLASCGSG FEGYITPPWSTRPAAEGQSWAVTGAQLEALKIDDVLIPYGLVFMWVEKDTV RTALDVMRKWGFYEAEGYSVTKRVNNTFTGTQRANGVFRSTKSLIUGRKK KKSEESGGDQLELRHQPNPDVNFERRKRFATGAAGVGERSEFMYRIETMLPKA RYDEETGTGKFLQLWNGYDMARPGWTVVCDTRTIRAPSAQHVV*	720
Asigmoides_jg4286.t1	6.1e-37	126.7	0.0	6mA DNA methyltransferase, RMSD score of 1.125	Asigmoides_jg4286.t1	MSDTGDRKKAAPRRQTTRVPFADSSDDEWVSCIDRLRGVSSLEGKIST HEAEAAAAKAAVAADAATAAAIAEAAAAAQAASSSSSSSSAAMDGTGA AAPGTMSSSSSAGSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSAAASPARDGATG LSAAQMASPSPHKTAPVATGGGGGADALDAALADLDIYEAPEWCVPV RADVRFDDWPKLAAAQFDVITMDPPWTLAGAAPTRGVAIGYQLSTRVIE ELPIPTLQTNGLAVWTINRFYAKAMEMMEGWGVEIDEVAVWKRVTNRKF AKGHGHLQHAREVCIGKGGTLPASIKAGTSDSVFAERRGQSQKPTIYIEM ERLAPNGRYLEIFARLNNLRDGVVWSIGNAL*	397
Asigmoides_jg9902.t1	9.8	3.9	7.5	-	Asigmoides_jg9902.t1	MAVPEPSRVLASLRGVRYTLIVRAVSLANLVIVRRVATGYVASQLLETLLV AALLVAREGTRRAALRVADLQVAAGGGAGGAGRAAGLGARGGKAAEKLVDG GQGLREKRRDQGRQEGREGQGVQGREETAAGDASMETVLGLGWATF FATCVLCALVGAVAAVAGAEARGMPGFTAAVVLTMA SCVAELAEPLFAA QHLEPARGVAVCAVLVRATATYALVARAGLGLVFAAGTLAHAVTLVACFA YSHIRASSGGSGHTDGDGRQKGPSTDLSTLDSTLDLPQVSTPSTRSL PRPLSALVPPSPARLATVLGGMPAETTLGLRGYTAQALLKVLSEGEKLLVARR PLHDQAVYSIVTSLGSIAARFLQPLEEAAVFSRLGGSGSGSSGSGSGNSG GSSKSGGGGGGNSGSGGGAGQASDALVRLVLLVRLVTHLGLFAAF GPPYARILLCLLYSESWACSDDAEVLGLYALLIPLALNLGTEAFHSTSTPRL LAMILNWWMLLGSVVTWAMLWHAADAGAVVVG MAGMSMRIAIVSVAVI AWRRRAAEYGFPRALADTVVAVLPHPAVALAFAICAAGAGSGQRWIEPVH GPVVHVAGAVIGVSLIAIIADPAFVRAIVSRRTIDKAKSE*	674

Figure 1. *Ancyromonas sigmoides* 6mA methyltransferase deep homology search table. Sequence ID shows the sequence ID pulled from the .txt file. E-value indicates the statistical significance of the sequence pulled. Score indicates how much the pulled sequence matches the MSA file. Bias indicates the bias score, which evaluates the tendency of MSA to over/under represent features. Description is to be filled over by the user after further analysis. FASTA Header show header pulled directly from FASTA. Sequence is the sequence put from FASTA, and length is counted by program.

The results of the 6mA methyltransferase search show 3 potential methyltransferase matches. The top match is jg4286.t1 with a e-value score of 6.1e-37, for e-value scores the smaller the value the higher the significance of the hit. Any e-value under e-10 is considered closely related. Jg3005.t1 has a slightly higher e-value of 1.3e-05 which may indicate a potential/weak relationship. Jg9902.t1 has the highest e-value which means it is the least likely to be a match, although anything under 10 is considered a weak match. Jg4286.t1 and jg3005.t1 both have bias scores of 0.0 which indicates MSA bias over or under representing features is unlikely to be an issue. Jg9902.t1 bias score of 7.5, might indicate it is a false positive, further investigation is needed. The sequences can be directly taken from this file and used for further methyltransferase investigation. For future directions I will take these sequences and run them in HHpred, then model them in Pymol with the HHpred top hits. With the e-value scores it is very likely that jg4286.t1 is a 6mA methyltransferase, indicating that *Ancyromonas*, although deep branching, share homologous 6mA methyltransferase enzymes.